

PRINCIPLES OF PROGG LANG CS F301

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Last Year's Data Abstraction and Object Orientation Tutorials

This is an assignment that is a follow-up on the first week assignment (a generalization of sorting), moving from procedural to OO proper. Indeed, do go back to your submissions of the first two weeks and review them, in order to do this better.

The concepts and questions for this are based on Chapter 10 of the textbook (the chapter heading is the same as this assignment title, this is Chapter 9 in the Third Edition, 10 in the subsequent two editions.). Attached also is a section of Stroustrup's book, "A Tour of C++" (Addison-Wesley Professional, 2018) on abstract types.

Chapter 10 contains development of a list class and suggestions for its generalization in two directions. One direction is making the data type behaviour (add, delete, update, rearrange--including but not only sorting) flexible and adaptable. Thus, a queue is a FIFO list, a stack is LIFO, and a deque (double-ended queue) allows addition and removal from either end without sequencing. A list also can be arranged in a heap and priority-queue topology. You need not go back to the tome of CLRS DSA textbook to read on these data types. All that you need to master is the design of the abstract type of a list-like container that admits the pertinent logical access methods (add, delete, update, rearrange or position a single element, maintaining the logical property of the chosen extension type) depending on the extension type (queue, stack, deque, priQ, heap as a priQ, etc.).

The other direction is allowing different types of elements, either in the same list or in different list instances without changing the list type.

Taking the code that Scott gives in 10.1 (and redesigning and developing it further) make

1. A list container as a storage structure with all the flexibilities and capabilities described above
2. A class hierarchy that implements each of the extensions described above

There will be a quiz on this design and development in due time (during one of the class or tutorial hours in the timetable, in LT1) that is evaluative, with 10% weightage of the total course marks. You have a week for the development and submission.



[Bjarne Stroustrup On Abstract Classes.pdf](#)2 August 2026, 8:50 PM

Add submission

Submission status

Attempt number	This is attempt 1 (4 attempts allowed).
Submission status	No submissions have been made yet
Grading status	Not graded

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